

[Dashboard](#)[my courses](#)[AMCN SS 2022](#)[Medium Access Schemes](#)[Quiz: Medium Access Schemes - TDMA \(Scheduling the Access\)](#)

Started at	Sunday, April 24, 2022 at 1:53 p.m
status	Completed
Finished at	Sunday, April 24, 2022, 2:07 p.m
Spent time	13 minutes 49 seconds
feedback	Congratulations, excellent performance (90-100%).



question 1

Complete

Achievable
points: 1.00

Compare the reserved-access and random access strategies with respect to the maximum load they can handle.

Choose one or more answers:

- ☒ a. Random access schemes exhibit lower maximum throughput due to high number of collisions
- ☒ b. Reservation-based schemes show higher network capacity due to the absence of data collision
- ☐ c. Random access schemes show higher capacity due to avoidance of overhead for reservation phase

question 2

Complete

Achievable
points: 1.00

How can the hidden terminal problem be solved?

Choose one or more answers:

- ☒ a. Use of short signaling packets to reserve for a single data transmission
- ☒ b. The receiver grants the right to transmit data to the sender
- ☒ c. Both sender and receiver inform their neighbors of an upcoming communication
- ☒ i.e. A sender initiates the transaction with a request of the right to send data to a receiver
- ☐ e. The sender grants the right to transmit data

question **3**

Complete

Achievable
points: 1.00

How does explicit reservation based on ALOHA work?

Wählen Sie eine oder mehrere Antworten:

- ☐ a. Stations compete during the beginning of the timeslot for it
- ☒ b. Stations compete during small reservation timeslots for data transmission timeslots
- ☒ c. The station needs to compete for the timeslot in each frame
- ☐ d. The station releases the timeslot only when it has no data to send

Frage **4**

Vollständig

Erreichbare
Punkte: 1,00

How does explicit reservation work?

Wählen Sie eine oder mehrere Antworten:

- ☐ a. Explicit reservation completely prevents collisions
- ☐ b. Explicit reservation requires a central controller
- ☐ c. Explicit reservation is used for longer terms only, e.g. a complete voice call
- ☒ d. The sender and receiver explicitly agree on a resource prior to its use



Frage 5

Vollständig

Erreichbare
Punkte: 1,00

How does implicit reservation work?

Wählen Sie eine oder mehrere Antworten:

- ☒ a. The station releases the timeslot implicitly when it does not have any more data to send
- ☒ b. After successful timeslot reservation, the station continues to use the same timeslot without competition in the following frame
- ☐ c. Implicit reservation prevents collisions on first access
- ☐ d. The station implicitly competes for the timeslot in each frame

Frage 6

Vollständig

Erreichbare
Punkte: 1,00

How does reservation TDMA work?

Wählen Sie eine oder mehrere Antworten:

- ☒ a. Each station is provided with a dedicated timeslot make a reservation
- ☒ b. A station may also compete for un-reserved data slots in each TDMA frame
- ☒ c. The mechanism ensures that each station has access to a guaranteed data volume in each frame
- ☐ d. Stations compete during small reservation timeslots for a data transmission timeslots
- ☒ e. Each stations is provided a small reservation timeslot to reserve a number of data slots in a collision free way



Frage 7

Vollständig

Erreichbare
Punkte: 1,00

Name the pros and cons of distributed approaches to media access?

Wählen Sie eine oder mehrere Antworten:

- ☐ a. Consistent view on the radio resources
- ☐ b. Not scalable
- ☒ c. No need for an infrastructure and management
- ☒ d. Inconsistent view on radio resource may result in collisions

Frage 8

Vollständig

Erreichbare
Punkte: 1,00

What are the pros and cons of the random access and reservation-based access strategies? Compare the access latencies of both access schemes.

Wählen Sie eine oder mehrere Antworten:

- ☐ a. The reservation-based access strategy has lower latency under low load
- ☐ b. The reservation-based strategy always experiences faster access under high load
- ☒ c. The random-access strategy has lower latency under low load
- ☒ d. The reservation-based access strategy may experience low latencies for data transmitted over given reservations



Frage 9

Vollständig

Erreichbare
Punkte: 1,00

What is correct for pure ALOHA?

Wählen Sie eine oder mehrere Antworten:

- ☐ a. It is a centralized scheme
- ☒ b. New packets are transmitted immediately
- ☐ c. New packets are transmitted after a random time
- ☒ d. In case of packet collisions, the station waits a random time and retransmits the packet again
- ☐ e. The station senses the medium before sending a packet

Frage 10

Vollständig

Erreichbare
Punkte: 1,00

When are collisions possible in MACA if we assume a stationary wireless system?

Wählen Sie eine oder mehrere Antworten:

- ☐ a. During data packet transmission
- ☒ b. During CTS packet transmission
- ☒ c. During RTS packet transmission

Frage **11**

Vollständig

Erreichbare
Punkte: 1,00

Which information is contained in signaling packets in MACA?

Wählen Sie eine oder mehrere Antworten:

- ☒ a. Packet size
- ☐ b. Frequency band
- ☒ c. Sender address
- ☐ d. Reservation time
- ☒ e. Receiver address

Frage **12**

Vollständig

Erreichbare
Punkte: 1,00

Which of the following problems are solved by RTS/CTS scheme (MACA)?

Wählen Sie eine oder mehrere Antworten:

- ☒ a. Exposed Terminal problem
- ☒ b. Hidden Terminal problem
- ☐ c. Near Far Terminal problem

Frage **13**

Vollständig

Erreichbare
Punkte: 1,00

Why does slotted ALOHA outperform pure ALOHA with respect to throughput?

Wählen Sie eine oder mehrere Antworten:

- ☐ a. In slotted ALOHA, each station competes in a short reservation slot to reserve a time slot
- ☒ b. In slotted ALOHA, two transmissions can collide only if they begin at the start of the same time slot
- ☐ c. Slotted ALOHA does not wait before transmitting

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